

Counts

A count has no upper bound, cannot be negative, and is nearly always more variable when it is large than when it is small. The linear model fits none of these descriptions and no transformation repairs all of them at once. The generalized linear model does better: put the linear predictor on the log scale, so that the mean is multiplicative, and let the distribution supply a variance that grows with the mean. That is the Poisson log-linear model, the starting point of this chapter.

It is only a starting point. Real counts are almost always more variable than the Poisson allows, often have far more zeros, and are sometimes not a sample of independent events at all but a contingency table in disguise. Each complaint has a model attached to it: exposure and rates, the exact connection between Poisson counts and multinomial proportions, the negative binomial, and the two standard ways to model an excess of zeros.

What you need. [Chapter 34](#) (the generalized linear model: Definition 34.2.1, the likelihood equations Theorem 34.3.1, iteratively reweighted least squares Theorem 34.4.1, the deviance Definition 34.5.1 and Theorem 34.5.5, and the three test statistics Proposition 34.5.3). [Chapter 35](#) supplies the separation theorem Theorem 35.3.1, whose Poisson counterpart is Proposition 37.1.3, and [Chapter 36](#) the baseline-category logit model Definition 36.1.1, which [Section 37.3](#) shows is a log-linear model in disguise. From earlier parts: the two-way layout and its interaction (Definition 16.2.1 and Proposition 16.2.1), variance-stabilizing transformations (Theorem 22.4.1), the sandwich covariance estimator (Theorem 21.4.2), the likelihood ratio test on the boundary of the parameter space (Theorem 32.5.2), and model selection by AIC and BIC (Definition 29.2.2).

37.-1.1 Roadmap

- [Poisson regression](#)
- [Offsets and rates](#)
- [Poisson and multinomial](#)
- [Negative binomial models](#)
- [Zero inflation](#)
- [Summary and notes](#)